Amine Maazizi

Research Intern at EPFL | Incoming Doctoral Researcher at Mines Paris – PSL

Foundation Models and Multimodal Learning for 3D Perception

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Research Profile

My research interests are representation learning, open-vocabulary 3D perception, and multimodal knowledge transfer from vision–language models to LiDAR and point clouds.

Education

Mines Paris – PSL — **Starting Nov. 2026**
Incoming PhD Student in Robotic Perception and 3D Machine Learning — Paris, France
Admitted to the doctoral programme; research begins in November 2026.

ENS Paris-Saclay — **2025 – 2026**
Research Master's in Mathematics, Vision and Learning (MVA) — Paris-Saclay, France
Highly selective graduate programme in machine learning and computer vision. Selected coursework: geometric deep learning, geometry processing, geometric data analysis, representation learning, graph machine learning, and multimodal explainable AI. Presented two research posters during the programme.

ENSTA Paris, Institut Polytechnique de Paris — **2023 – 2026**
Engineering Degree in Computer Science and Artificial Intelligence — Palaiseau, France
French *Diplôme d'Ingénieur*, a master's-level engineering qualification. GPA: **4.05/4.30**; ranked in the top 5%. Selected coursework: statistical learning, machine learning, deep learning, computer vision, language models, reinforcement learning.

Lycée Moulay Youssef — **2021 – 2023**
Preparatory Classes in Mathematics and Physics (MPSI/MP)* — Rabat, Morocco
Two-year intensive CPGE programme, tracks MPSI/MP*, with coursework in mathematics, physics, and computer science; ranked in the top 5% nationally.

Research Experience

Mines Paris – PSL, Centre for Robotics (CAOR) — **Starting Nov. 2026**
Incoming Doctoral Researcher, NPM3D – Point Cloud and 3D Modeling — France
Supervisors: Prof. Jean-Emmanuel Deschaud, Prof. François Goulette, and Prof. Philippe Xu

- PhD project on transferring semantic knowledge from vision–language foundation models to LiDAR and point-cloud representations for autonomous navigation in natural environments.
- Planned research directions include open-vocabulary 3D perception, cross-modal knowledge transfer, and robust semantic understanding of unstructured outdoor scenes.

EPFL, Artificial Intelligence in Molecular Medicine Laboratory (AIMM) — **Apr. 2026 – Sep. 2026**
Research Intern — Lausanne, Switzerland
Supervisors: Prof. Charlotte Bunne and Dr. Lukas Klein

- Developing conditional perturbation models to predict patient-specific treatment responses in spatial proteomics using representations from the VirTues foundation model.
- Investigating distributional modelling approaches, including flow matching and conditional reweighting, for mapping pre-treatment tissue states to post-treatment cell-state distributions.
- Designing evaluation protocols for distributional fidelity, biological response recovery, patient-identity preservation, and conditioning effectiveness.

Institut Pasteur, Biological Image Analysis Unit — **Jun. 2025 – Dec. 2025**
Research Intern and Continuing Collaborator — Paris, France
Supervisor: Prof. Giacomo Nardi

- Developed a geometry-aware shape-matching framework for biological meshes, combining geometric descriptors, constrained correspondence, and mechanically motivated regularization.
- Applied the framework to the morphodynamic analysis of the endothelial-to-hematopoietic transition and implemented the associated modelling and evaluation pipeline.

NAIST, Cybernetics and Reality Engineering Laboratory (CaRE) — **Oct. 2024 – Dec. 2024**
Research Collaborator — Nara, Japan
Supervisor: Prof. Hideaki Uchiyama

- Conducted a structured study of text- and image-conditioned 3D generation, analysing major methodological families, evaluation practices, and limitations in geometric consistency and controllability.

Selected Research Outputs and Software

not-MIWAE PyTorch — **2026**
Open-source research software — GitHub / PyPI
PyTorch implementation of MIWAE, not-MIWAE, supMIWAE, and not-supMIWAE within a unified probabilistic framework for learning from missing-not-at-random data.

PointLLM Contrastive Alignment — **2026**
Multimodal 3D representation-learning project — GitHub
Investigated contrastive objectives for point-cloud–language alignment in a PointLLM-style architecture and compared them with generative language-modelling objectives.

Geometry-Aware Shape Matching — **2025 – 2026**
Research software — Private repository
Modular mesh-correspondence framework combining geometric descriptors, constrained matching, and mechanically motivated regularization for morphodynamic analysis of biological shapes.

Posters

A Critical Review of Zero-Shot Super-Resolution for Preclinical MRI — **2026**
A. Maazizi *and N. Vujadinovic* — MVA research poster
Critical review and lightweight reproduction of diffusion-based biplanar super-resolution methods for anisotropic preclinical MRI, presented for the MVA course *Deep Learning for Medical Imaging*.

Deep Generative Modelling with MNAR Data and a Supervised Extension — **2026**
A. Gassem, ***A. Maazizi****, and E. Melzani* — MVA research poster
Reproduced MIWAE, supMIWAE, and not-MIWAE, introduced a supervised MNAR extension, evaluated the models on image data, and released an open-source PyTorch implementation, presented for the MVA course *Introduction to Probabilistic Graphical Models and Deep Generative Models*

Awards and Honors

Finalist, Best Research Project Award, ENSTA Paris — **2025**
Moroccan Excellence Scholarship, Ministry of Higher Education — **2023**

Technical Skills

Machine Learning:	PyTorch, TensorFlow, scikit-learn, Hugging Face Transformers, Hugging Face Datasets.
3D & Scientific:	Open3D, PyVista, PyTorch Geometric, NumPy, SciPy.
Software & Systems:	Python, C, C++, Linux, Git, Docker, SLURM, Run:ai.

Languages

Arabic (native) | French (bilingual) | English (fluent) | Spanish (elementary)